

AZ-104: Microsoft Azure Administrator - Complete Study Guide

Exam Overview

As a candidate for this exam, you should have subject matter expertise in implementing, managing, and monitoring an organization's Microsoft Azure environment, including virtual networks, storage, compute, identity, security, and governance.

The exam covers five main domains:

- Manage Azure identities and governance (20-25%)
- Implement and manage storage (15-20%)
- Deploy and manage Azure compute resources (20-25%)
- Implement and manage virtual networking (15-20%)
- Monitor and maintain Azure resources (10-15%)

A score of 700 or greater is required to pass.

DOMAIN 1: Manage Azure Identities and Governance (20-25%)

1.1 Manage Microsoft Entra ID Users and Groups

Understanding Microsoft Entra ID (formerly Azure AD)

Microsoft Entra ID is Azure's cloud-based identity and access management solution.

Microsoft Entra ID includes features like comparing Microsoft Entra ID to Active Directory DS, learning about Microsoft Entra ID P1 and P2, and Microsoft Entra Domain Services for managing domain-joined devices and apps in the cloud.

A tenant is a specific instance of Microsoft Entra ID containing accounts and groups.

Creating and Managing Users

User accounts will store user data such as name, department, location, and contact information.

User Types:

- **Cloud identities:** Users created directly in Microsoft Entra ID
- **Synchronized identities:** Users synced from on-premises AD using Microsoft Entra Connect
- **Guest users:** External users invited via B2B collaboration

Key User Properties:

- User Principal Name (UPN)
- Display name
- Job title
- Department
- Usage location (required for license assignment)

Managing Groups

Group accounts can include user accounts or devices. These are two basic ways members are assigned to groups: Statically and Dynamically. Static groups require administrators to add and remove members manually. Dynamic groups update automatically based on the properties of a user account or device. For example, job title.

Group Types:

- **Security groups:** Used to manage member and computer access to shared resources
- **Microsoft 365 groups:** Provide collaboration opportunities with shared mailbox, calendar, files

Membership Types:

- **Assigned:** Manually add/remove members
- **Dynamic User:** Automatically add/remove users based on attribute rules
- **Dynamic Device:** Automatically add/remove devices based on attribute rules (Security groups only)

An Entra ID Premium P1 or P2 license is required for dynamic membership.

Expiration lets you configure a group lifetime in days. After that time the group must be renewed by the owner. Naming policy lets you configure blocked words and add a prefix or suffix to group names.

Managing Licenses

- Licenses can be assigned directly to users or through group-based licensing
- Usage location must be set before assigning licenses
- License assignment options:

- Direct assignment to individual users
- Group-based licensing (requires P1 or P2)

Self-Service Password Reset (SSPR)

SSPR Configuration:

- Can be enabled for all users, selected groups, or none
- Requires at least one authentication method
- Available methods: Mobile phone, email, security questions, Microsoft Authenticator app
- Writeback to on-premises AD requires Azure AD Connect

SSPR Requirements:

- Users must be registered for SSPR
- Administrator accounts always require two authentication methods
- Security questions not available for administrators

1.2 Manage Access to Azure Resources

Azure Role-Based Access Control (RBAC)

Azure role-based access control (Azure RBAC) helps you manage who has access to Azure resources, what they can do with those resources, and what areas they have access to. Azure RBAC is an authorization system built on Azure Resource Manager.

The way you control access to resources using Azure RBAC is to assign Azure roles. This is a key concept to understand – it's how permissions are enforced. A role assignment consists of three elements: security principal, role definition, and scope. A security principal is an object that represents a user, group, service principal, or managed identity that is requesting access to Azure resources. You can assign a role to any of these security principals.

A role definition is a collection of permissions. It's typically just called a role. A role definition lists the actions that can be performed, such as read, write, and delete. Roles can be high-level, like owner, or specific, like virtual machine reader. Azure includes several built-in roles that you can use. For example, the Virtual Machine Contributor role allows a user to create and manage virtual machines.

Fundamental Azure Roles:

1. **Owner:** Full access to all resources, can delegate access to others
2. **Contributor:** Can create and manage all types of Azure resources, cannot grant access
3. **Reader:** Can view existing Azure resources
4. **User Access Administrator:** Can manage user access to Azure resources

RBAC Scope Levels (hierarchical):

- Management Group
- Subscription
- Resource Group
- Individual Resource

Azure RBAC is an additive model, so your effective permissions are the sum of your role assignments.

Creating Custom Roles:

- Custom roles can be created when built-in roles don't meet needs
- Defined using JSON with Actions, NotActions, DataActions, NotDataActions
- Can be scoped to management group, subscription, or resource group

If the built-in roles don't meet the specific needs of your organization, you can create your own Azure custom roles.

Resource Locks

Lock Types:

- **CanNotDelete (Delete lock):** Authorized users can read and modify, but cannot delete
- **ReadOnly (Read-only lock):** Authorized users can read, but cannot modify or delete

Key Concepts:

- Locks apply regardless of RBAC permissions
- Locks are inherited by child resources
- Only Owner and User Access Administrator can create/delete locks
- Must remove lock before performing restricted operation

1.3 Manage Azure Subscriptions and Governance

Management Groups

If your organization has many Azure subscriptions, you might need a way to efficiently manage access, policies, and compliance for those subscriptions. Management groups provide a governance scope above subscriptions. When you organize subscriptions into management groups, the governance conditions that you apply cascade by inheritance to all associated subscriptions. Management groups give you enterprise-grade management at scale, no matter what type of subscriptions you might have.

Another scenario where you would use management groups is to provide user access to multiple subscriptions. By moving multiple subscriptions under a management group, you can create one Azure role assignment on the management group. The role inherits that access to all the

subscriptions. One assignment on the management group can enable users to have access to everything they need, instead of scripting Azure role-based access control (RBAC) over different subscriptions.

Key Points:

- Root management group exists by default for each directory
- Can nest up to 6 levels deep
- Each management group can have many children
- Each child can have only one parent
- Policies and RBAC assignments cascade down

Azure Policy

Azure Policy evaluates resources and actions in Azure by comparing the properties of those resources to business rules. These business rules, described in JSON format, are known as policy definitions. To simplify management, several business rules can be grouped together to form a policy initiative, also referred to as a policySet.

After your business rules are formed, the policy definition or initiative is assigned to any scope of resources that Azure supports. For example, management groups, subscriptions, resource groups, or individual resources. The assignment applies to all resources within the Resource Manager scope of that assignment.

Policy Effects:

- **Audit:** Creates a warning event in the activity log
- **AuditIfNotExists:** Audits if related resource doesn't exist
- **Deny:** Denies the resource request
- **DeployIfNotExists:** Deploys a resource if it doesn't exist
- **Disabled:** Policy is turned off
- **Modify:** Add, update, or remove properties or tags

Policy Initiative:

With an initiative definition, you can group several policy definitions to achieve one overarching goal. An initiative evaluates resources within scope of the assignment for compliance to the included policies.

Resource Tags

Tag Concepts:

- Name-value pairs applied to resources, resource groups, and subscriptions
- Up to 50 tags per resource/resource group/subscription
- Tag names are case-insensitive; tag values are case-sensitive

- Not inherited by default (can use Policy to enforce inheritance)

Common Tag Uses:

- Cost management and billing
- Operations management
- Security and compliance
- Governance and policy compliance
- Automation and workload optimization

Cost Management

Azure Cost Management Features:

- **Budgets:** Set spending limits with alerts
- **Cost Analysis:** View and analyze current costs
- **Cost Alerts:** Notifications when spending thresholds are reached
- **Azure Advisor Recommendations:** Suggestions to optimize costs

Budget Alert Types:

- Actual cost alerts: Trigger when actual spending exceeds threshold
- Forecasted cost alerts: Trigger when projected spending will exceed threshold

DOMAIN 2: Implement and Manage Storage (15-20%)

2.1 Configure Access to Storage

Storage Account Types

The following table describes the types of storage accounts recommended by Microsoft for most scenarios. All of these use the Azure Resource Manager deployment model.

Storage Account Types:

Type	Supported Services	Performance Tiers	Use Cases
Standard general-purpose v2	Blob, Queue, Table, File, Data Lake	Standard	Most scenarios
Premium block blobs	Blob (including Data Lake)	Premium	High transaction scenarios
Premium file shares	Azure Files	Premium	Enterprise file shares

Type	Supported Services	Performance Tiers	Use Cases
Premium page blobs	Page blobs only	Premium	VM disks, databases

Storage Account Endpoints

A storage account provides a unique namespace in Azure for your data. Every object that you store in Azure Storage has a URL address that includes your unique account name. The combination of the account name and the service endpoint forms the endpoints for your storage account.

Default Endpoints:

- Blob: <https://<storage-account>.blob.core.windows.net>
- File: <https://<storage-account>.file.core.windows.net>
- Queue: <https://<storage-account>.queue.core.windows.net>
- Table: <https://<storage-account>.table.core.windows.net>
- Data Lake Storage Gen2: <https://<storage-account>.dfs.core.windows.net>

Storage Redundancy Options

Azure Storage always stores multiple copies of your data to protect it from planned and unplanned events. Examples of these events include transient hardware failures, network or power outages, and massive natural disasters. Redundancy ensures that your storage account meets its availability and durability targets even in the face of failures.

Primary Region Redundancy:

Locally redundant storage (LRS) replicates the data within your storage accounts to one or more Azure availability zones located in the primary region of your choice. LRS provides at least 99.999999999% (11 nines) durability of objects over a given year. LRS is the lowest-cost redundancy option and offers the least durability compared to other options. LRS protects your data against drive, server, and rack failures. However, if a disaster such as fire or flooding occurs within the data center, all replicas of a storage account using LRS might be lost or unrecoverable.

Zone-redundant storage (ZRS) copies your data synchronously across three or more Azure availability zones in the primary region.

Geo-Redundancy Options:

Option	Description	Secondary Read Access
GRS	LRS in primary + async copy to secondary region	No
RA-GRS	LRS in primary + readable secondary	Yes

Option	Description	Secondary Read Access
GZRS	ZRS in primary + async copy to secondary region	No
RA-GZRS	ZRS in primary + readable secondary	Yes

Data in a GZRS account is copied across three or more Azure availability zones in the primary region. In addition, it also replicates to a secondary geographic region for protection from regional disasters. Microsoft recommends using GZRS for applications requiring maximum consistency, durability, and availability, excellent performance, and resilience for disaster recovery. With a GZRS account, you can continue to read and write data if an availability zone becomes unavailable or is unrecoverable. Additionally, your data also remains durable during a complete regional outage or a disaster in which the primary region isn't recoverable. GZRS is designed to provide at least 99.99999999999999% (16 9s) durability of objects over a given year.

Shared Access Signatures (SAS)

SAS Types:

1. **Account SAS:** Access to resources in one or more storage services
2. **Service SAS:** Access to resources in a single storage service
3. **User delegation SAS:** Secured with Microsoft Entra credentials (Blob storage only)

SAS Components:

- Signed resource (blob, container, etc.)
- Permissions (read, write, delete, list)
- Start and expiry time
- IP address restrictions
- Protocol (HTTPS only or HTTP/HTTPS)
- Signature

Best Practices:

- Use HTTPS only
- Use User Delegation SAS when possible
- Set expiry times as short as practical
- Apply principle of least privilege

Storage Firewalls and Virtual Networks

Network Access Configuration:

- **Allow all networks:** Default, all traffic allowed
- **Allow selected virtual networks and IP addresses:** Restrict to specific networks
- **Disabled:** Block all public network access

Private Endpoints:

- Create private IP address in your VNet for storage
- Traffic stays on Microsoft backbone network
- Can disable public endpoint access entirely

2.2 Configure and Manage Storage Accounts

Blob Storage Access Tiers

Azure storage offers different access tiers so that you can store your blob data in the most cost-effective manner based on how it's being used.

Access Tiers:

Tier	Use Case	Storage Cost	Access Cost	Min Retention
Hot	Frequently accessed data	Higher	Lower	None
Cool	Infrequently accessed (30+ days)	Lower	Higher	30 days
Cold	Rarely accessed (90+ days)	Lower than Cool	Higher than Cool	90 days
Archive	Long-term backup/compliance	Lowest	Highest	180 days

The hot, cool, and cold tiers support all redundancy configurations. Only storage accounts that are configured for LRS, GRS, or RA-GRS support moving blobs to the archive tier. The archive tier isn't supported for ZRS, GZRS, or RA-GZRS accounts.

Lifecycle Management

Blob storage lifecycle management offers a rule-based policy that you can use to transition your data to the desired access tier when your specified conditions are met. You can also use lifecycle management to expire data at the end of its life.

Lifecycle management policies are free of charge. Customers are billed for standard operation costs for the Set Blob Tier API calls. Delete operations are free.

Lifecycle Policy Actions:

- `tierToCool` : Move to cool tier
- `tierToCold` : Move to cold tier
- `tierToArchive` : Move to archive tier
- `delete` : Delete blobs
- `enableAutoTierToHotFromCool` : Auto-move back to hot when accessed

Lifecycle Policy Conditions:

- `daysAfterModificationGreaterThan`
- `daysAfterCreationGreaterThan`
- `daysAfterLastAccessTimeGreaterThan` (requires access time tracking)

Object Replication

Object Replication Concepts:

- Asynchronously copies block blobs between source and destination accounts
- Requires blob versioning on both accounts
- Requires change feed on source account
- Doesn't support snapshots

Use Cases:

- Minimize latency by placing data closer to users
- Increase efficiency for compute workloads
- Optimize data distribution
- Optimize costs by moving replicated data to archive tier

2.3 Configure Azure Files and Azure Blob Storage

Azure Files

Azure Files Features:

- SMB and NFS protocol support
- Fully managed file shares
- Can be mounted on Windows, Linux, macOS
- Can be cached on Windows Servers using Azure File Sync

File Share Types:

- **Standard:** HDD-based, transaction-optimized
- **Premium:** SSD-based, high IOPS and throughput

Authentication Methods:

- Identity-based (Azure AD DS, on-premises AD DS, Microsoft Entra Kerberos)
- Shared key (access keys)

Azure File Sync

Components:

- **Storage Sync Service:** Top-level Azure resource
- **Sync Group:** Defines sync topology for files
- **Registered Server:** Trust relationship between server and Storage Sync Service
- **Azure File Sync Agent:** Installed on Windows Server
- **Server Endpoint:** Location on registered server that syncs
- **Cloud Endpoint:** Azure file share

Cloud Tiering:

- Frequently accessed files cached locally
- Infrequently accessed files tiered to cloud
- Policies based on volume free space and date

Blob Storage

Blob Types:

- **Block blobs:** Store text and binary data (up to ~190.7 TiB)
- **Append blobs:** Optimized for append operations (logs)
- **Page blobs:** Random access files (VHDs) up to 8 TiB

Blob Containers:

- Organize blobs within storage account
- Unlimited blobs per container
- Access levels: Private, Blob (anonymous read for blobs), Container (anonymous read for container and blobs)

Blob Versioning and Soft Delete:

- Blob versioning: Maintain previous versions automatically
- Soft delete: Retain deleted data for specified period
- Container soft delete: Protect containers from deletion

DOMAIN 3: Deploy and Manage Azure Compute Resources (20-25%)

3.1 Automate Deployment Using ARM Templates or Bicep

Azure Resource Manager (ARM) Templates

Template Components:

```
{
  "$schema": "...",
  "contentVersion": "1.0.0.0",
  "parameters": { },
  "variables": { },
  "resources": [ ],
  "outputs": { }
}
```

Key Sections:

- **\$schema**: Location of JSON schema file
- **contentVersion**: Version of template
- **parameters**: Values provided during deployment
- **variables**: Reusable values in template
- **resources**: Resources to deploy or update
- **outputs**: Values returned after deployment

Bicep

Bicep Features:

- Domain-specific language (DSL) for deploying Azure resources
- Simpler syntax than ARM templates
- Compiles to ARM template JSON
- First-class support in Azure

Bicep vs ARM Template:

- Cleaner syntax without JSON complexity
- Automatic dependency management
- Type safety and IntelliSense support
- Modules for reusability

3.2 Create and Configure Virtual Machines

VM Sizing

VM Series:

Series	Use Case
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Series	Use Case
A-series	Entry-level, dev/test
B-series	Burstable workloads
D-series	General purpose
E-series	Memory optimized
F-series	Compute optimized
G-series	Memory and storage optimized
H-series	High-performance computing
L-series	Storage optimized
M-series	Memory optimized (large)
N-series	GPU-enabled

VM Storage

Disk Types:

Type	Description	Use Case
Ultra Disk	Highest IOPS/throughput	IO-intensive workloads
Premium SSD v2	Flexible IOPS/throughput	Production workloads
Premium SSD	Guaranteed IOPS	Production workloads
Standard SSD	Consistent performance	Web servers, dev/test
Standard HDD	Lowest cost	Backup, infrequent access

Disk Roles:

- **OS Disk:** Contains operating system (max 4095 GiB)
- **Temporary Disk:** Short-term storage (data lost on reboot)
- **Data Disks:** Application data, can attach multiple

VM Availability

Availability zones expands the level of control you have to maintain the availability of the applications and data on your VMs. An Availability Zone is a physically separate zone, within an Azure region. There are three Availability Zones per supported Azure region. Each Availability Zone has a distinct power source, network, and cooling. By designing your solutions to use replicated VMs in zones, you can protect your apps and data from the loss of a data center. If one zone is compromised, then replicated apps and data are instantly available in another zone.

Availability Sets:

Availability sets are logical groupings of VMs that reduce the chance of correlated failures bringing down related VMs at the same time. Availability sets place VMs in different fault domains for better reliability.

The underlying Azure platform assigns an update domain and a fault domain to each virtual machine in your availability set. Each availability set can have up to 3 fault domains and 20 update domains.

Update domains indicate groups of virtual machines and underlying physical hardware that can be restarted at the same time.

Fault domains define the group of virtual machines that share a common power source and network switch. By default, the virtual machines configured within your availability set are separated across up to three fault domains. Placing your virtual machines into an availability set doesn't protect your application from operating system or application-specific failures. But it does limit the impact of potential physical hardware failures, network outages, or power interruptions.

VM Extensions

Common Extensions:

- Custom Script Extension: Run scripts on VMs
- Desired State Configuration (DSC): Configure VMs
- Azure Diagnostics: Collect diagnostic data
- Network Watcher Agent: Network monitoring
- Log Analytics Agent: Send data to Log Analytics

3.3 Provision and Manage Containers

Azure Container Instances (ACI)

ACI Features:

- Fast startup (seconds)
- Per-second billing
- Custom sizes (CPU, memory)
- Persistent storage with Azure Files
- Linux and Windows containers

Container Groups:

- Multiple containers on same host
- Share lifecycle, network, storage
- Similar to Kubernetes pod concept

- Scheduled on same host machine

Azure Container Apps

Container Apps Features:

- Serverless containers
- Automatic scaling (including to zero)
- Built-in KEDA support
- Dapr integration
- Revision management
- Traffic splitting

3.4 Create and Configure Azure App Service

App Service Plans

Each tier also provides a specific subset of App Service features. These features include custom domains and TLS/SSL certificates, autoscaling, deployment slots, backups, Azure Traffic Manager integration, and more. The higher the tier, the more features that are available. To find out which features are supported in each pricing tier, see the App Service plan details. You can find more comparisons of plans in App Service limits.

App Service Plan Tiers:

Tier	Features
Free/Shared	Shared infrastructure, no SLA
Basic	Dedicated compute, manual scale
Standard	Autoscale, deployment slots, backups
Premium	More instances, staging slots, VNet integration
Isolated	Dedicated VNet, highest scale

Deployment Slots

When you deploy your web app, web app on Linux, mobile back end, or API app to Azure App Service, you can use a separate deployment slot instead of the default production slot. This approach is available if you run in the Standard, Premium, or Isolated tier of an App Service plan. Deployment slots are live apps with their own host names. App content and configuration elements can be swapped between two deployment slots, including the production slot.

Deploying your application to a nonproduction slot has the following benefits: You can validate app changes before you swap the slot into production. You can make sure that all instances of the slot are warmed up before you swap it into production. This approach eliminates downtime when you deploy your app. The traffic redirection is seamless. No requests are dropped because of swap operations. You can automate this entire workflow by configuring auto swap when pre-swap validation isn't needed.

The number of deployment slots available depends on the pricing tier of your App Service plan: Standard Tier: Up to 5 deployment slots per app. Premium and PremiumV2 Tiers: Up to 20 deployment slots per app.

Slot-Specific Settings (not swapped):

- Publishing endpoints
- Custom domain names
- Non-public certificates and TLS/SSL settings
- Scale settings
- WebJobs schedulers
- IP restrictions
- Always On
- Diagnostic settings
- Cross-origin resource sharing (CORS)

App Service Scaling

Scale Up (Vertical):

- Change App Service plan tier
- More CPU, memory, disk space
- Additional features

Scale Out (Horizontal):

- Increase number of instances
- Manual or automatic scaling
- Based on metrics (CPU, memory, HTTP queue, etc.)

Autoscale Rules:

- Metric-based: Scale based on performance metrics
- Schedule-based: Scale on specific schedules
- Combine multiple rules with AND/OR logic

3.5 Virtual Machine Scale Sets

Azure virtual machine scale sets let you create and manage a group of load balanced VMs. The number of VM instances can automatically increase or decrease in response to demand or a defined schedule. Scale sets provide high availability to your applications, and allow you to centrally manage, configure, and update many VMs. There is no cost for the scale set itself, you only pay for each VM instance that you create. Virtual machines in a scale set can also be deployed into multiple availability zones, a single availability zone, or regionally.

Scale sets support the use of the Azure load balancer for basic layer-4 traffic distribution, and Azure Application Gateway for more advanced layer-7 traffic distribution and TLS termination.

Orchestration Modes:

- **Flexible:** Full control over VMs, mix VM sizes and types
- **Uniform:** Optimized for large-scale workloads with identical VMs

Scaling Options:

- Manual scaling: Set instance count directly
- Autoscaling: Based on metrics or schedules
- Predictive autoscaling: Uses ML to anticipate demand

Update Policies:

- **Automatic:** Updates applied immediately
 - **Rolling:** Updates in batches
 - **Manual:** Updates applied when triggered
-

DOMAIN 4: Implement and Manage Virtual Networking (15-20%)

4.1 Configure and Manage Virtual Networks

Virtual Network Concepts

In Azure, a VNet is the fundamental building block that allows virtual machines, applications, and other services to communicate safely with each other, with the internet, and with on-premises networks.

A virtual network in Azure is a logical construct that mirrors an on-premises network, yet it exists entirely in the cloud. Each Azure VNet is private and dedicated, with IP addresses drawn from RFC 1918 private ranges (for example, 10.0.0.0/8).

VNet Components:

- Address space (CIDR notation)
- Subnets
- Network security groups
- Route tables
- Service endpoints
- Private endpoints

Subnets

Subnet Reserved Addresses:

For each subnet, Azure reserves 5 IP addresses:

- x.x.x.0: Network address
- x.x.x.1: Default gateway
- x.x.x.2, x.x.x.3: Azure DNS mapping
- x.x.x.255: Broadcast address

Special Subnets:

- GatewaySubnet: For VPN/ExpressRoute gateways (/27 minimum recommended)
- AzureFirewallSubnet: For Azure Firewall (/26 minimum)
- AzureBastionSubnet: For Azure Bastion (/26 minimum)

Virtual Network Peering

No downtime to resources in either virtual network when you create the peering or after the peering is created. Network traffic between peered virtual networks is private. Traffic between the virtual networks is kept on the Microsoft backbone network. No public internet, gateways, or encryption are required in the communication between the virtual networks.

Peering Types:

- **Virtual network peering:** Same Azure region
- **Global virtual network peering:** Across Azure regions

Gateway transit is a peering property that enables a virtual network to use a virtual private network or an ExpressRoute gateway in a peered virtual network. Gateway transit works for both cross-premises and network-to-network connectivity. Traffic to the gateway (ingress or egress) in the peered virtual network incurs virtual network peering charges on the spoke virtual network.

Peering Requirements:

- Non-overlapping IP address spaces

- Cannot be changed after creation
- Peering is non-transitive (requires hub-spoke or mesh)

4.2 Configure Secure Access to Virtual Networks

Network Security Groups (NSGs)

NSG Components:

- Security rules with priority (100-4096, lower = higher priority)
- Direction: Inbound or Outbound
- Source/Destination: IP addresses, service tags, ASGs
- Protocol: TCP, UDP, ICMP, or Any
- Port range
- Action: Allow or Deny

Default Rules:

- AllowVNetInBound/OutBound
- AllowAzureLoadBalancerInBound
- DenyAllInBound/OutBound

Service Tags:

- VirtualNetwork
- Internet
- AzureLoadBalancer
- Storage
- Sql
- And many more Azure service-specific tags

Application Security Groups (ASGs)

ASG Benefits:

- Group NICs with similar functions
- Use as source/destination in NSG rules
- No need to manage individual IP addresses
- Simplify complex rule management

Azure Firewall

Features:

- Stateful firewall as a service

- Built-in high availability
- Threat intelligence-based filtering
- FQDN filtering
- Network and application rules
- SNAT and DNAT support

Azure Bastion

Features:

- Secure RDP/SSH over SSL
- No public IP on target VMs required
- Protection against port scanning
- Integrated with Azure portal

4.3 Configure Name Resolution and Load Balancing

Azure DNS

Public DNS Zones:

- Host domain records in Azure
- Support for common record types (A, AAAA, CNAME, MX, TXT, etc.)
- Alias record sets for Azure resources
- DNSSEC not currently supported

Private DNS Zones:

- Name resolution within virtual networks
- Automatic VM registration
- Split-horizon DNS support
- Can link multiple VNets

DNS Record Types:

Type	Description
A	Maps hostname to IPv4 address
AAAA	Maps hostname to IPv6 address
CNAME	Canonical name (alias)
MX	Mail exchange
NS	Name server

Type	Description
PTR	Pointer (reverse DNS)
SOA	Start of authority
SRV	Service locator
TXT	Text record

Azure Load Balancer

Azure Load Balancer distributes traffic between multiple virtual machines.

Load Balancer SKUs:

Feature	Basic	Standard
Backend pool size	Up to 300	Up to 1000
Health probes	TCP, HTTP	TCP, HTTP, HTTPS
Availability zones	No	Yes
SLA	None	99.99%
Secure by default	No	Yes

Load Balancer Components:

- Frontend IP configuration
- Backend pools
- Health probes
- Load balancing rules
- Inbound NAT rules

Distribution Modes:

- Five-tuple hash (default): Source IP, source port, destination IP, destination port, protocol
- Session persistence: Source IP affinity (two-tuple or three-tuple)

Application Gateway

Azure Application Gateway provides advanced layer-7 traffic distribution and TLS termination.

Features:

- URL-based routing
- Multi-site hosting
- SSL/TLS termination

- Web Application Firewall (WAF)
- Autoscaling
- Zone redundancy
- Session affinity

4.4 Configure VPN Gateway and ExpressRoute

VPN Gateway

VPN Types:

- **Site-to-Site (S2S):** Connect on-premises network to Azure VNet
- **Point-to-Site (P2S):** Connect individual computers to Azure VNet
- **VNet-to-VNet:** Connect Azure VNets together

VPN Gateway SKUs:

SKU	S2S Tunnels	P2S Connections	Aggregate Throughput
Basic	10	128	100 Mbps
VpnGw1	30	250	650 Mbps
VpnGw2	30	500	1 Gbps
VpnGw3	30	1000	1.25 Gbps
VpnGw4	100	5000	5 Gbps
VpnGw5	100	10000	10 Gbps

The gateway subnet must be /27 or a shorter prefix, such as /26 or /25, or you receive an error message when you add the ExpressRoute virtual network gateway.

ExpressRoute

Azure ExpressRoute. You can use ExpressRoute to extend your on-premises networks into the Microsoft cloud over a private connection, with the help of a connectivity provider. With ExpressRoute, you can establish connections to Microsoft cloud services, like Azure and Microsoft 365.

ExpressRoute Peering Types:

- **Private peering:** Connect to VNets via private IPs
- **Microsoft peering:** Connect to Microsoft 365 and Azure public services

ExpressRoute Models:

- CloudExchange co-location
- Point-to-point Ethernet connection
- Any-to-any (IPVPN) connection
- ExpressRoute Direct

You can configure a site-to-site VPN connection as a backup for your ExpressRoute connection. This setup applies only to virtual networks linked to the Azure private peering path. There's no VPN-based failover solution for services accessible through Azure Microsoft peering. The ExpressRoute circuit is always the primary link, and data flows through the site-to-site VPN path only if the ExpressRoute circuit fails.

DOMAIN 5: Monitor and Maintain Azure Resources (10-15%)

5.1 Monitor Resources in Azure

Azure Monitor

Azure Monitor collects and aggregates metrics and logs from your system to monitor availability, performance, and resilience, and notify you of issues affecting your system. You can use the Azure portal, PowerShell, Azure CLI, REST API, or client libraries to set up and view monitoring data. Different metrics and logs are available for different resource types.

Data Types:

- **Metrics:** Numerical values at regular intervals (near real-time)
- **Logs:** Detailed information stored in Log Analytics workspace
- **Activity Logs:** Subscription-level events

Azure Monitor Components:

- Metrics Explorer
- Log Analytics
- Application Insights
- Alerts
- Insights (VM Insights, Container Insights, etc.)

Log Analytics

Log Analytics allows you to query and analyze log data using the Kusto query language (KQL).

Log Analytics Workspace:

- Central repository for log data
- Data collected from multiple sources
- Query data using KQL
- Retention period: 30-730 days

Common Data Sources:

- Azure resources (diagnostics settings)
- Virtual machines (agents)
- Applications (Application Insights)
- Custom logs

Azure Alerts

Azure Monitor alerts allow you to identify and address issues in your system, and proactively notify you when specific conditions are found in your monitoring data before your customers notice them.

Metric alerts evaluate resource metrics at regular intervals. Metrics can be platform metrics, custom metrics, logs from Azure Monitor converted to metrics, or Application Insights metrics. Metric alerts can also apply multiple conditions and dynamic thresholds. Log alerts allow users to use a Log Analytics query to evaluate resource logs at a predefined frequency. Activity log alerts trigger when a new activity log event occurs that matches defined conditions.

Alert Components:

- **Alert rules:** Conditions that trigger alerts
- **Action groups:** Actions to take when alert fires
- **Alert processing rules:** Modify alert behavior (suppress, add actions)

Action Group Actions:

- Email/SMS/Push/Voice
- Azure Function
- Logic App
- Webhook
- ITSM
- Automation Runbook

Alert Severity Levels:

Severity	Name	Description
Sev 0	Critical	Immediate action required
Sev 1	Error	Urgent attention needed
Sev 2	Warning	Attention needed

Severity	Name	Description
Sev 3	Informational	Useful information
Sev 4	Verbose	Detailed diagnostic info

Azure Monitor Insights

VM Insights:

- Performance monitoring
- Dependency mapping
- Health monitoring
- Guest OS metrics

Container Insights:

- Kubernetes cluster monitoring
- Container performance
- Node health
- Logs collection

Network Insights:

- Network topology
- Connectivity monitoring
- Traffic analytics

5.2 Implement Backup and Recovery

Azure Backup

Azure Backup offers integration with various Azure services, such as Azure Resource Graph, Azure Monitor Alerts, Azure Monitor Logs (Log Analytics), and Azure Resource Health. Azure Backup also offers interfaces to various Azure clients, such as Azure portal, Azure PowerShell, Azure CLI, and Azure REST API. Depending on your monitoring and reporting requirements, you can use a combination of these integrations.

Backup Components:

Recovery Services Vault:

- Central management of backup data
- Backup policies
- Soft delete protection (14 days by default)
- Cross-region restore (for GRS vaults)

Supported Workloads:

- Azure VMs (Windows and Linux)
- Azure Files
- SQL Server in Azure VMs
- SAP HANA in Azure VMs
- On-premises (MARS agent, DPM, MABS)
- Azure Blobs
- Azure Managed Disks

Backup Policy Components:

- Schedule (daily, weekly, etc.)
- Retention (daily, weekly, monthly, yearly points)
- Instant restore snapshots

Recovery Options:

- Create new VM
- Replace existing disk
- Restore files

Azure Site Recovery

Capabilities:

- Disaster recovery for Azure VMs
- On-premises to Azure replication
- VMware/Hyper-V to Azure
- Orchestrated failover/failback

Replication Policy Settings:

- Recovery point retention
- App-consistent snapshot frequency
- Replication frequency

Recovery Plans:

- Define failover order
- Add manual actions
- Run scripts during failover
- Group machines together

VM Backup Best Practices

Pre-backup:

- Verify VM agent is running
- Check backup extension status
- Ensure sufficient disk space

Retention Considerations:

- Balance retention with cost
- Use GRS for geo-redundancy
- Consider cross-region restore needs

Restore Considerations:

- Test restores regularly
 - Document restore procedures
 - Understand RTO/RPO requirements
-

Quick Reference: PowerShell and Azure CLI Commands

Resource Groups

```
# PowerShell
New-AzResourceGroup -Name "RG-Name" -Location "eastus"
Get-AzResourceGroup -Name "RG-Name"
Remove-AzResourceGroup -Name "RG-Name"

# Azure CLI
az group create --name RG-Name --location eastus
az group show --name RG-Name
az group delete --name RG-Name
```

Virtual Machines

```
# PowerShell
New-AzVM -ResourceGroupName "RG" -Name "VM" -Location "eastus"
Get-AzVM -ResourceGroupName "RG" -Name "VM"
Stop-AzVM -ResourceGroupName "RG" -Name "VM"
Start-AzVM -ResourceGroupName "RG" -Name "VM"
```

```
# Azure CLI
az vm create --resource-group RG --name VM --image UbuntuLTS
az vm show --resource-group RG --name VM
az vm stop --resource-group RG --name VM
az vm start --resource-group RG --name VM
```

Storage Accounts

```
# PowerShell
New-AzStorageAccount -ResourceGroupName "RG" -Name "storageaccount" -Location "eastus"
Get-AzStorageAccount -ResourceGroupName "RG" -Name "storageaccount"

# Azure CLI
az storage account create --name storageaccount --resource-group RG --location eastus
az storage account show --name storageaccount --resource-group RG
```

Virtual Networks

```
# PowerShell
New-AzVirtualNetwork -Name "VNet" -ResourceGroupName "RG" -Location "eastus"
Add-AzVirtualNetworkSubnetConfig -Name "Subnet" -VirtualNetwork $vnet -AddressPrefix 10.0.0.0/24

# Azure CLI
az network vnet create --name VNet --resource-group RG --address-prefix 10.0.0.0/24
az network vnet subnet create --vnet-name VNet --resource-group RG --name Subnet --address-prefix 10.0.0.0/24
```

Exam Tips

1. **Hands-on Practice:** Use the Azure free account to practice in the portal and with CLI/PowerShell
2. **Know the Limits:** Understand key service limits (NSG rules, storage accounts per subscription, etc.)
3. **Understand Pricing Tiers:** Know which features require which tiers (e.g., deployment slots require Standard+)
4. **Focus on High-Weight Topics:** Compute and Identity/Governance each account for 20-25%

5. **Practice Scenarios:** The exam includes case studies and scenario-based questions
 6. **Know Both Portal and CLI:** Be familiar with performing tasks through both interfaces
 7. **Understand Dependencies:** Know what's required before creating resources (e.g., VNet before VM, resource group before anything)
 8. **Review Microsoft Learn Paths:** Complete the official AZ-104 learning paths on Microsoft Learn
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